Aakash Yadav E-mail: aakashyadav1098642@gmail.com
LinkedIn: www.linkedin.com/in/aakash-yadav-56931b28a Mobile: 8318806831
GitHub: https://github.com/SOUL665

EDUCATION

- Vellore Institute of Technology (Bhopal) — Bhopal, MP
  B.Tech in Computer Science and Engineering — Current CGPA: 7.98
  (Artificial Intelligence and Machine Learning)

- Kalpanath Inter College, Ghazipur — Ghazipur,UP
  Higher Secondary Education — Percentage: 83.3

SKILLS SUMMARY

- **Tech Skills:** Java, Python, C++ (Basic), AWS , Generative AI Tools
- **Soft Skills:** Teamwork, Adaptability, Fast Learner, Time Management, Communication Skills
- **Languages:** English, Hindi
- **Hobbies:** Chess, Reading, Watching Anime

PROJECTS

- ***Pothole Detection System (Python, YOLOv8, Deep Learning)***
  - Developed an end-to-end automated pothole detection system by fine-tuning a pretrained YOLOv8n model on ~1,977 annotated road images using transfer learning, achieving a mAP50 of 0.715 and Precision of 0.74.
  - Implemented a detection pipeline supporting both static images and video footage, achieving ~14 FPS inference speed on CPU without any GPU acceleration.
  - Built and deployed an interactive Streamlit web application enabling non-technical users to upload road images and receive instant pothole detection results with bounding boxes and pothole count.
  - GitHub Link: https://github.com/SOUL665/Pothole-Detector-Project.git

- ***Matrix Cipher Encoder/Decoder (Python)***
  - Developed a command-line tool to encode and decode messages using the Hill cipher (matrix-based classical encryption).
  - Implemented matrix operations (including modular arithmetic and matrix inversion) using NumPy.
  - Designed a user-friendly interface for key input, message padding, and error handling.
  - Demonstrated understanding of cryptographic concepts and Python programming.
  - GitHub Link: https://github.com/SOUL665/Matrix-Chiper.git

CERTIFICATES

- Java Foundations Associate |Oracle [View Certificate]
- Applied Machine Learning in Python | University of Michigan [View Certificate]
- GEN AI using IBM Watson | IBM [View Certificate]
- Cloud Computing | NPTEL [View Certificate]

Co-Curricular Activities

- **Team Member, VIT Chess League 5**
  Secured 2nd place as part of an 8-member team in the 2024 edition of the college chess league.

- **Member, GeeksForGeeks Student Chapter Club**
  Participated in organizing and attending club activities and events